

APIA FALEOLO, Samoa

WMO: 917590

Lat: 13.830S

Lon: 172.008W

Elev: 58

StdP: 14.67

Time Zone: 13.00 (PCS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	69.7	71.2	65.7	94.9	75.0	66.8	98.7	75.0	24.0	84.9	21.6	84.5	2.5	180	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.2	90.0	79.7	89.7	79.6	89.2	79.4	81.7	87.4	81.0	87.0	80.6	86.8	12.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.2	157.0	85.6	79.2	151.7	84.8	79.0	150.6	84.6	45.4	88.0	44.8	87.9	44.3	86.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.9	17.2	66.7	92.4	2.3	1.0	65.1	93.1	63.8	93.6	62.5	94.2	60.9	94.9
			64.6	84.0	2.9	1.6	62.6	85.2	60.9	86.1	59.3	87.0	57.2	88.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	81.7	81.8	81.8	81.7	81.0	80.4	79.5	79.5	80.0	80.4	81.0	81.3
	DBStd	1.78	1.74	1.64	1.48	1.39	1.55	1.55	1.76	1.57	1.57	1.51	1.48	1.60
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11248	982	891	986	950	959	912	913	913	901	942	930	969
	CDD65	5773	517	471	521	500	494	462	448	448	451	477	480	504
	CDH74	55485	5352	4897	5365	4987	4655	4175	3810	3870	4166	4502	4684	5022
	CDH80	18325	1756	1630	1843	1725	1529	1351	1184	1243	1415	1501	1529	1619
Wind	WSAvg	5.9	7.1	6.6	5.6	4.9	4.6	5.6	5.9	6.2	6.7	6.1	5.5	5.7
Precipitation	PrecAvg	83.3	11.8	9.8	8.2	6.7	6.6	3.5	3.7	3.3	4.5	6.7	8.7	9.7
	PrecMax	99.1	24.8	23.9	16.1	21.1	14.8	8.5	8.6	7.9	10.8	12.3	15.7	14.9
	PrecMin	57.8	4.8	2.2	1.9	1.5	0.6	1.3	1.1	0.5	0.6	2.8	4.4	4.2
	PrecStd	8.9	4.9	5.2	3.7	4.1	3.1	1.5	1.9	1.7	2.2	2.2	2.8	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	91.0	91.1	91.2	90.0	89.9	89.7	89.8	90.2	90.0	90.0	90.2
		MCWB	80.5	80.5	80.5	80.5	80.0	78.8	78.3	78.5	78.7	79.0	79.4	80.2
	2%	DB	89.5	89.4	89.6	89.6	89.2	88.2	88.1	88.1	89.1	89.3	89.2	89.1
		MCWB	80.1	80.0	80.0	79.9	79.7	78.7	78.0	77.9	78.2	78.8	79.3	79.7
	5%	DB	88.1	88.1	88.2	88.2	87.9	87.7	87.3	87.3	87.8	87.8	87.8	87.9
		MCWB	79.8	79.7	79.7	79.7	79.6	78.6	77.7	77.6	78.0	78.4	79.0	79.3
	10%	DB	87.2	87.4	87.6	87.6	86.9	86.3	86.0	86.0	86.3	86.3	86.4	86.6
		MCWB	79.5	79.5	79.5	79.5	79.0	77.9	77.4	77.1	77.4	77.9	78.4	78.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	81.9	81.9	81.8	82.1	81.0	80.8	80.1	81.0	81.4	81.3	81.8
		MCDB	87.2	87.0	87.9	87.8	87.3	87.0	87.2	86.9	87.3	87.7	87.6	88.0
	2%	WB	81.0	81.0	81.0	80.9	80.7	80.0	79.6	79.2	79.6	80.0	80.1	80.7
		MCDB	86.7	86.7	87.1	87.2	87.0	86.3	86.0	85.9	86.4	86.5	86.5	86.7
	5%	WB	80.4	80.3	80.3	80.1	80.0	79.3	78.8	78.4	78.9	79.2	79.6	79.9
		MCDB	86.2	86.1	86.4	86.3	86.1	85.6	85.0	85.0	85.7	85.6	85.8	85.7
	10%	WB	79.8	79.7	79.7	79.7	79.3	78.6	78.0	77.8	78.1	78.5	79.0	79.2
		MCDB	85.3	85.3	85.6	85.7	85.3	84.8	84.2	84.3	84.8	84.7	85.1	84.8
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.2	12.1	12.6	12.7	13.2	13.4	13.8	13.7	13.2	12.3	11.7
		MCDBR	12.6	12.5	13.3	13.6	13.9	14.5	15.4	15.5	15.2	14.9	13.6	13.2
		MCWBR	5.8	5.8	6.3	6.6	6.8	7.1	7.6	7.6	7.3	7.4	6.6	6.2
	5% WB	MCDBR	11.6	11.6	12.5	12.8	13.0	13.4	13.6	14.1	14.0	13.3	12.6	12.1
MCWBR		5.7	5.8	6.1	6.5	6.7	7.0	7.4	7.4	7.4	7.3	6.5	6.2	
Clear-Sky Solar Irradiance	taub	0.386	0.378	0.375	0.382	0.391	0.389	0.389	0.391	0.394	0.397	0.387	0.385	0.385
	taud	2.626	2.658	2.668	2.641	2.597	2.588	2.576	2.576	2.558	2.552	2.595	2.614	2.623
	Ebn at Noon	304	304	298	286	273	268	270	278	287	297	302	304	304
	Edh at Noon	32	31	30	29	29	28	29	31	33	33	32	32	32
All-Sky Solar Radiation	RadAvg	1810	1806	1812	1630	1442	1373	1421	1646	1844	1905	1871	1844	1844
	RadStd	178	171	132	132	99	71	59	78	103	96	99	113	113

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page